THE COTANGENT COMPLEX OF A SCHEME OVER A RING IS THE COTANGENT COMPLEX OF SHEAVES OF RINGS

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file cotangent.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The statement displays the identification and names the constant sheaf; the proof is a three-step reduction ending in a localization vanishing.

Editorial note. The three references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: the retained preamble of this chapter (sources/stacks-cotangent/retained-preamble.tex, lines 1-63), containing the Xy-pic setup, the comment environments, the theorem declarations and the macros including \NL.

Allowed standard prerequisites (editorial).

- The distinguished triangle for a composition of morphisms of ringed spaces. Lemma lemma-triangle-ringed-spaces (3763-3771, produced as L44162), used at proof lines 4181-4182. Reduces the statement to the vanishing of L_q.
- The description of the stalk of the cotangent complex. Lemma lemma-stalk-cotangent-complex, used at proof lines 4183-4189. Turns the vanishing into a statement about a local ring.
- The vanishing of the cotangent complex of a localization. Lemma lemma-when-zero (1832-1849, produced as L44166), used at proof lines 4189-4190. Gives the required vanishing.
- The definition of the cotangent complex of a morphism of schemes. used at proof lines 4169-4175. Fixes the objects appearing in the identification.

Source reference. Original lemma 08T0: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11148–11148. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 08T4: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11152–11152. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 08R2: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11080–11080. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *Let Λ be a ring. Let X be a scheme over Λ. Then*

$$L_{X/\operatorname{Spec}(\Lambda)} = L_{\mathcal{O}_X/\underline{\Lambda}}$$

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where $\underline{\Lambda}$ is the constant sheaf with value Λ on X.

Proof. Let $p : X \to \mathrm{Spec}(\Lambda)$ be the structure morphism. Let $q : \mathrm{Spec}(\Lambda) \to (*, \Lambda)$ be the obvious morphism. By the distinguished triangle of Lemma 08T4 it suffices to show that $L_q = 0$. To see this it suffices to show for $\mathfrak{p} \in \mathrm{Spec}(\Lambda)$ that

$$(L_q)_{\mathfrak{p}} = L\mathcal{O}_{\mathrm{Spec}(\Lambda),\mathfrak{p}}/\Lambda = L_{\Lambda_{\mathfrak{p}}/\Lambda}$$

(Lemma 08T0) is zero which follows from Lemma 08R2. □